

GENERAL THETA FOUNDATIONS I: FOURIER BUDGETS FOR HIDDEN CAUSAL MEMORY

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ABSTRACT. We prove a polynomial memory lower bound for a uniformly positive rotation experiment whose separate positive minimum is three at every cut. The bound allows arbitrary hidden states, unrelated transition matrices at different epochs, and a new machine for each horizon. Its proof uses conditional past-phase moments rather than the number of vertices of a projected simplex section. A homogeneous angular inequality controls all Fourier moments by one cumulative loss of conditional phase amplitude; positivity of the Fejér kernel detects a register with few labels. For an angle satisfying $\inf_{m \geq 1} m \|m\alpha\| > 0$, the unrestricted width is at least a constant times $N^{1/5}$, improving the previous logarithmic lower bound, while an exact $O_\alpha(N^{1/3})$ construction is retained. The lower bound persists under any fixed sufficiently small uniform output error. For the rational rotation $(3 + 4i)/5$ it gives a logarithmic, rather than doubly logarithmic, lower bound. We formulate stochastic extension as finite Blackwell comparison and transfer the new obstruction to the accumulated deficiency of every proposed lift diagram. This rules out logarithmic-coordinate low-defect lift chains for the bounded-type family without imposing vector-state or equivariance restrictions. The exponents $1/5$ and $1/3$ are not identified, and the fixed-length and anytime autonomous specifications are kept distinct.

1. INTRODUCTION

An experiment can admit a small positive factorization at each separate cut while requiring substantially more states when the factorizations must be joined by common stochastic updates. For hidden realizations, a state need not carry an individually meaningful predictive vector. A projection of a K -coordinate simplex section can have exponentially many vertices. Consequently, a lower bound obtained only by counting the vertices of an observable polygon may lose an exponential factor in the memory parameter.

We study the following fixed-alphabet experiment. A seed $x \in \{-1, 1\}^2$ is followed by N commands $c_t \in \{0, 1\}$, a query $j \in \{1, 2\}$, and a binary answer B . For $\zeta = e^{2\pi i \alpha}$ and the corresponding planar rotation R_α , the prescribed answer law is

$$(1) \quad \mathbb{P}(B = b \mid x, c_1, \dots, c_N, j) = \frac{1 + b e_j^T R_\alpha^{c_1 + \dots + c_N} x / 10}{2}.$$

All these probabilities lie in the fixed interval $[(1 - \sqrt{2}/10)/2, (1 + \sqrt{2}/10)/2]$. The alphabets, amplitude and support margin do not change with N . Exact realization means equality for every input word, not merely after averaging over commands.

The separate positive minimum of this experiment is three. The previous mean-support argument gave $\Omega_\alpha(\log N)$ as an unrestricted hidden-state lower bound and $\Theta_\alpha(N^{1/3})$ within a vector-state subclass at bounded-type angles [17]. We retain its exact upper construction but replace the hidden lower argument. The new method does not associate a predictive vector with each basis state. Instead, it studies the conditional expectation of a random past phase *under a*

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chosen test distribution on external words. These conditional expectations are proof objects, not information made available to the executing machine.

Write

$$(2) \quad \beta_m(\alpha) = \min_{1 \leq l \leq m} \sin^2(\pi l \alpha), \quad b_\alpha = \inf_{l \geq 1} l \|l \alpha\|_{\mathbb{R}/\mathbb{Z}}.$$

For irrational α , every β_m is positive. The assumption $b_\alpha > 0$ is the usual badly-approximable, equivalently bounded-type, condition. Let $W_{N,\varepsilon}(\alpha)$ be the least peak label count of a clocked stochastic realization with uniform row-total-variation error at most ε , and write $W_N = W_{N,0}$.

Theorem 1.1 (Polynomial obstruction for all hidden realizations). *Let $0 \leq \varepsilon < 1/20$ and put $\kappa = 1/10 - 2\varepsilon$. Every realizing profile $(K_t)_{t=0}^N$ satisfies, for every integer $k \geq 1$,*

$$(3) \quad \#\{0 \leq t < N : K_t \leq k\} \beta_{2k}(\alpha) \leq \frac{18k^3}{\kappa^2}.$$

In particular, if $b_\alpha > 0$, then

$$(4) \quad W_{N,\varepsilon}(\alpha) \geq \left(\frac{b_\alpha^2 \kappa^2 N}{18} \right)^{1/5}.$$

For $\alpha = (\sqrt{5} - 1)/2$, exact realization therefore obeys

$$(5) \quad \max\{3, (N/16200)^{1/5}\} \leq W_N(\alpha) < 2 \max\{5, (40N)^{1/3}\}.$$

More generally the same cube-root upper order holds whenever the partial quotients of α are bounded. For $\zeta = (3 + 4i)/5$,

$$(6) \quad N \leq 7200 W_N^3 25^{2W_N}, \quad W_N \geq \max\{3, \log_{5000}(N/7200)\}.$$

All lower bounds allow arbitrary cut-dependent hidden-state continuations.

The theorem excludes $O(\log N)$ hidden width at bounded-type angles, including at any fixed row error below $1/20$. It does not prove that hidden and vector-state widths have the same order: the current exact bounds are $\Omega_\alpha(N^{1/5})$ and $O_\alpha(N^{1/3})$. The cumulative version (3) also limits how often a small register may occur in an otherwise unrestricted profile.

The proof has three steps. Conditional phase amplitude decreases under a stochastic update. Its entire loss is at most one. A homogeneous angular inequality bounds the perturbation of the l th Fourier moment by $2l^2$ times this same loss. Finally, the positive Fejér kernel forces a nonnegligible Fourier potential whenever the conditional phase measure has few atoms. Randomizing the two external commands damps those moments at the rates $\sin^2(\pi l \alpha)$. The resulting potential telescopes for time-dependent as well as stationary matrices.

Section 5 interprets a supplied lift diagram as a sequence of finite experiment comparisons. The one-step garbling criterion and deficiency dual are classical [3, 12]. Their composition produces an executable approximate transducer. Applying Theorem 1.1 yields a new lower bound on the total defect of *every* such diagram for (1), rather than only a four-point warning example. Section 6 compares this notion with equivariant polyhedral lifts and distinguishes fixed-length autonomy from the stronger anytime query task.

Notation	Meaning
$K_t, W_{N,\varepsilon}$	Available labels at cut t ; minimum peak at row error ε .
$z_t(s), q_t$	Conditional past-phase expectation; its mean absolute value.
$\Phi_t^{(m)}$	Finite Fejér-weighted Fourier potential of the conditional phases.
$\delta(E, F)$	One-sided finite deficiency: simulate F by garbling E .
$\mathcal{D}(\mathcal{E})$	Sum of optimized one-step and terminal defects for a fixed diagram.
$A_N^{\text{fix}}, A_N^{\text{any}}$	Autonomous state counts for one fixed length, or all lengths up to N .

2. THE FINITE STOCHASTIC MODEL

State distributions are row vectors. Let $\Delta_k = \{e \in \mathbb{R}^k : e \geq 0, e\mathbf{1} = 1\}$.

Definition 2.1 (Clocked transducer and approximation). A transducer for a prescribed N has initialization rows $E_x \in \Delta_{K_0}$, row-stochastic matrices $T_{t,c} \in \mathbb{R}_+^{K_t \times K_{t+1}}$ for $0 \leq t < N$, $c \in \{0, 1\}$, and binary decoder means $d_j \in [-1, 1]^{K_N}$ for $j \in \{1, 2\}$. Its mean after a word is

$$E_x T_{0,c_1} \cdots T_{N-1,c_N} d_j.$$

Its uniform row error is the maximum total variation between this binary law and (1), over all seeds, command words and queries. The available-label profile is (K_t) and its peak is $\max_t K_t$. $W_{N,\varepsilon}$ minimizes this peak over transducers of row error at most ε . At error zero the defining product must equal the specified mean on every word.

Only the current state survives each update. All retained randomness, past inputs and protocol distinctions belong to that state. The seed and query are supplied once as atomic symbols. The command is an external input; it is not a hidden parameter chosen by the model. The epoch is an external clock, so different cuts can have different alphabets and rows. All rows, including unreachable rows, are normalized. Available labels are counted; padding is allowed. A retained final binary answer uses two labels at its own output cut. This does not enlarge any exact peak here, which is at least three.

Exact stochastic rows, their read-only descriptions, arithmetic, and row-construction time are not charged by this positive-realization width. A rational row can be compiled into fair-bit rejection sampling with extra phase, old-state, current-input and sampler-prefix storage, as in [15]. No equality between that implementation cost and the atomic width is asserted. Irrational rows below are exact mathematical kernels, not a finite fair-bit implementation claim.

Proposition 2.2 (The separate positive minimum remains three). *For exact realization of (1), every internal cut has normalized residual rank three and separate positive minimum three. The separate minimum allows arbitrary finite registers at all other cuts.*

Proof. A prefix has a predictive vector $R_\alpha^a x/10$. Its complete future array is affine in that vector. Four idle prefixes span an affine plane, and two idle-suffix coordinate queries separate it. Normalization adds the constant direction, so the residual rank is three. The triangle with vertices

$$(1/2, 0), \quad (-1/2, 1/2), \quad (-1/2, -1/2)$$

contains the disk of radius $\sqrt{2}/10$. Indeed its supporting-line distances from zero are $1/2, 1/(2\sqrt{5}), 1/(2\sqrt{5})$, all larger than that radius. Its vertices have norm at most $1/\sqrt{2}$, so every rotated continuation from a vertex gives legal binary answers. At the selected cut decompose the actual predictive vector in this triangle. Before that cut store the finite raw history; after it store the chosen vertex label and remaining command count. This is a complete finite realization with three states at the selected cut. Rank proves minimality. The construction is inherited from [16, 17]; the new result concerns simultaneous, not separate, width. $\square$

The lower argument below uses only the stochastic update rule and the correct terminal correlation. It does not project hidden basis states into this predictive plane. In particular, it covers the simplex-section lifts in [17], including lifts with unobservable directions.

3. A FOURIER BUDGET FOR A HIDDEN REGISTER

This section states the probabilistic argument independently of the particular seed amplitude. Let $C_1, \dots, C_N$ be independent fair bits, put $L_t = C_1 + \cdots + C_t$, and let S_t be an arbitrary

finite-state process with

$$(7) \quad \mathbb{P}(S_{t+1} = s' \mid S_0, C_1, \dots, C_t, S_t, C_{t+1}) = T_{t, C_{t+1}}(S_t, s').$$

The new command is independent of the past. Initialization may be random. The matrices and state spaces may depend on t and N . Set $Y_t = \zeta^{-L_t}$ and, on states of positive probability, define

$$(8) \quad p_t(s) = \mathbb{P}(S_t = s), \quad z_t(s) = \mathbb{E}[Y_t \mid S_t = s], \quad q_t = \sum_s p_t(s) |z_t(s)|.$$

Put $z_t = 0$ on a zero-mass state. These are conditional expectations used in the proof, not registers or extra inputs to the transducer.

Lemma 3.1 (One common amplitude loss). *The sequence q_t is nonincreasing, with $q_0 = 1$. If $Z_{t+1} = \zeta^{-C_{t+1}} z_t(S_t)$, then*

$$(9) \quad z_{t+1}(S_{t+1}) = \mathbb{E}[Z_{t+1} \mid S_{t+1}], \quad \ell_t := q_t - q_{t+1} \geq 0, \quad \sum_{t < N} \ell_t \leq 1.$$

If a complex function $d(S_N)$ obeys $|d| \leq B$ and $|\mathbb{E}[Y_N d(S_N)]| \geq B\kappa$, then $q_t \geq \kappa$ for every t .

Proof. The stochastic update and the independence of the new command imply $\mathbb{E}[Y_t \mid S_t, C_{t+1}, S_{t+1}] = z_t(S_t)$ on positive-mass events. The tower property gives the first identity. Conditional Jensen gives $q_{t+1} \leq \mathbb{E}|Z_{t+1}| = q_t$, hence the telescoping loss bound. Finally

$$|\mathbb{E}[Y_N d(S_N)]| = |\mathbb{E}[z_N(S_N) d(S_N)]| \leq Bq_N.$$

Monotonicity proves the last assertion. No conditional independence of past commands given the retained state is assumed. $\square$

For an integer $l \geq 1$ define the homogeneous angular function

$$(10) \quad f_l(re^{i\theta}) = re^{il\theta} \quad (r > 0), \quad f_l(0) = 0.$$

It is not necessary to differentiate this function at zero. The following inequality charges its failure to preserve averages to the *same* absolute-amplitude loss for every l .

Lemma 3.2 (Homogeneous angular averaging). *For any integrable complex random variable Z and integer $l \geq 1$,*

$$(11) \quad |\mathbb{E}f_l(Z) - f_l(\mathbb{E}Z)| \leq (l^2 + 1)(\mathbb{E}|Z| - |\mathbb{E}Z|).$$

Proof. If $\mathbb{E}Z = 0$, the left side is at most $\mathbb{E}|Z|$, and the result follows. Otherwise rotate so that $\mathbb{E}Z = \rho > 0$, and write $Z = re^{i\theta}$. Then $\mathbb{E}(r \sin \theta) = 0$ and $\mathbb{E}(r \cos \theta) = \rho$. For $u = e^{i\theta}$ the identity

$$u^l - 1 - l(u - 1) = (u - 1) \sum_{j=0}^{l-1} (u^j - 1)$$

and $|u^j - 1| \leq j|u - 1|$ give

$$|u^l - 1 - il \sin \theta| \leq l^2(1 - \cos \theta).$$

Consequently

$$|e^{il\theta} - \cos \theta - il \sin \theta| \leq (l^2 + 1)(1 - \cos \theta).$$

Multiply by r , average, and use the two identities for sine and cosine. Rotation covariance of f_l restores the general case. Values with $r = 0$ contribute zero. Thus the proof covers mixtures passing through zero. $\square$

Let

$$(12) \quad m_{t,l} = \sum_s p_t(s) f_l(z_t(s)), \quad a_l = \frac{1 + \zeta^{-l}}{2}.$$

By radial homogeneity, $|m_{t,l}| \leq q_t \leq 1$.

Lemma 3.3 (Moment recursion). *There are complex errors $e_{t,l}$ such that*

$$(13) \quad m_{t+1,l} = a_l m_{t,l} + e_{t,l}, \quad |e_{t,l}| \leq (l^2 + 1) \ell_t \leq 2l^2 \ell_t.$$

In particular

$$(14) \quad |m_{t+1,l}|^2 \leq (1 - \sin^2(\pi l \alpha)) |m_{t,l}|^2 + 4l^2 \ell_t.$$

Proof. Apply Lemma 3.2 conditionally on S_{t+1} to Z_{t+1} in Lemma 3.1, then average. The total conditional amplitude deficit is exactly ℓ_t . Rotation covariance and the fair command give $\mathbb{E} f_l(Z_{t+1}) = a_l m_{t,l}$. This proves the first line. Both $|m_{t+1,l}|$ and $|a_l m_{t,l}|$ are at most one, so the difference of their squares is at most $2|e_{t,l}|$. Use $|a_l|^2 = 1 - \sin^2(\pi l \alpha)$. $\square$

Associate to cut t the positive measure on the circle

$$(15) \quad \nu_t = \sum_{s: z_t(s) \neq 0} p_t(s) |z_t(s)| \delta_{\arg z_t(s)}.$$

Its mass is q_t , it has at most K_t atoms, and its l th Fourier moment is $m_{t,l}$. Equal directions may be combined; doing so only reduces the number of atoms.

Lemma 3.4 (A finite positive Fourier witness). *For a positive circle measure ν of mass q with at most k atoms, put*

$$\Phi^{(M)}(\nu) = \sum_{l=1}^M \left(1 - \frac{l}{M+1}\right) |\hat{\nu}(l)|^2.$$

Then

$$(16) \quad q^2 + 2\Phi^{(M)}(\nu) \geq \frac{M+1}{k} q^2.$$

In particular, $\Phi^{(2k)}(\nu) \geq q^2/2$.

Proof. The Fejér kernel

$$F_M(\theta) = \frac{1}{M+1} \left| \sum_{j=0}^M e^{ij\theta} \right|^2 = 1 + 2 \sum_{l=1}^M \left(1 - \frac{l}{M+1}\right) \cos(l\theta)$$

is nonnegative and has value $M+1$ at zero. If the atom weights are w_i , integrate $F_M(\theta - \phi)$ against $\nu \otimes \nu$. Dropping the nonnegative off-diagonal terms gives at least $(M+1) \sum_i w_i^2 \geq (M+1) q^2/k$. Expanding the same integral gives the left side of (16). This is the elementary positive-kernel identity; its use here is to detect a finite hidden register. $\square$

Theorem 3.5 (Spectral occupation budget). *In the process (7), suppose $q_N \geq \kappa > 0$. For every integer $k \geq 1$, even if other cuts have arbitrarily many states,*

$$(17) \quad \#\{0 \leq t < N : K_t \leq k\} \beta_{2k}(\alpha) \leq \frac{18k^3}{\kappa^2}.$$

The conclusion remains true with K_t replaced by the number of positive-probability states under this test distribution.

Proof. Fix k , set $M = 2k$, $w_l = 1 - l/(M + 1)$, and abbreviate $\Phi_t = \Phi^{(M)}(\nu_t)$. Multiply (14) by w_l and sum. This gives

$$\Phi_{t+1} \leq \Phi_t - \sum_{l=1}^M w_l \sin^2(\pi l \alpha) |m_{t,l}|^2 + 4\ell_t \sum_{l=1}^M w_l l^2.$$

Initially $z_0 = 1$ on every positive-mass state, so $m_{0,l} = 1$ and $\Phi_0 = M/2 = k$. Moreover

$$\sum_{l=1}^M w_l l^2 = \frac{M(M+1)(M+2)}{12} = \frac{k(k+1)(2k+1)}{3} \leq 2k^3.$$

Sum over all epochs, use $\Phi_N \geq 0$ and $\sum \ell_t \leq 1$, and obtain

$$(18) \quad \sum_{t < N} \sum_{l=1}^{2k} w_l \sin^2(\pi l \alpha) |m_{t,l}|^2 \leq k + 8k^3 \leq 9k^3.$$

At any cut with at most k positive-mass states, Lemma 3.4 and $q_t \geq q_N \geq \kappa$ imply $\Phi_t \geq \kappa^2/2$. The summand at such a cut is therefore at least $\beta_{2k}(\alpha)\kappa^2/2$. All other summands are nonnegative. Counting the small cuts proves (17). Nothing in the proof identifies the conditional past phase with a statewise future prediction, and no bound on vertices of a lift is used. $\square$

The test law uses fresh independent commands but makes no such assumption about the commands in the specification: a wordwise simulator must pass every chosen test distribution. The conditional objects in (8) need not be computable from the finite state by an available runtime table; they are used only in the converse.

4. CONSEQUENCES FOR THE ROTATION EXPERIMENT

We now apply the budget to the original wordwise specification, not to a statewise-intertwining subclass.

Proposition 4.1 (Terminal calibration of the past phase). *A realization of (1) with uniform row-TV error at most $\varepsilon < 1/20$ satisfies $q_N \geq 1/10 - 2\varepsilon$ under the fair-command test law with fixed seed $x = (1, 1)$.*

Proof. Let $d = d_1 + id_2$ be the complex conditional mean of the two possible answers from the final state. Its modulus is at most $\sqrt{2}$. For any fixed command word, exactness would give

$$\mathbb{E}[d(S_N) \mid C_1, \dots, C_N] = \frac{1+i}{10} \zeta^{L_N}.$$

A TV error ε for a binary answer changes its mean by at most 2ε . The complex error in this display is therefore at most $2\sqrt{2}\varepsilon$. Multiply by $Y_N = \zeta^{-L_N}$ and average. The reverse triangle inequality gives

$$|\mathbb{E}[Y_N d(S_N)]| \geq \sqrt{2}(1/10 - 2\varepsilon).$$

Apply Lemma 3.1. Only the final-length answer identities were used. There is no assumption of a correct answer at an earlier cut. $\square$

Proof of the lower assertions of Theorem 1.1. Proposition 4.1 and Theorem 3.5 give (3). If every cut has at most K states, then $N\beta_{2K}(\alpha) \leq 18K^3/\kappa^2$. For $b_\alpha > 0$,

$$|\sin(\pi l \alpha)| \geq 2\|l\alpha\|_{\mathbb{R}/\mathbb{Z}} \geq \frac{2b_\alpha}{l} \geq \frac{b_\alpha}{K} \quad (1 \leq l \leq 2K).$$

This proves (4).

For the golden angle set $\alpha' = -(1 + \sqrt{5})/2$. If p is an integer nearest $l\alpha$, then

$$(l\alpha - p)(l\alpha' - p) = p^2 + lp - l^2 \in \mathbb{Z} \setminus \{0\}, \quad |l\alpha' - p| \leq \sqrt{5}l + 1/2 < 3l.$$

Hence $l\|\alpha\| > 1/3$ and $b_\alpha \geq 1/3$. With $\kappa = 1/10$ this gives $(N/16200)^{1/5}$. The exact rank-three bound is Proposition 2.2.

For $\zeta = (3 + 4i)/5$, no positive power equals one: otherwise $\zeta + \zeta^{-1} = 6/5$ would be a rational algebraic integer. Thus $(3 + 4i)^l - 5^l$ is a nonzero Gaussian integer and has modulus at least one. It follows that

$$\sin^2(\pi l \alpha) = \frac{|\zeta^l - 1|^2}{4} \geq \frac{25^{-l}}{4}, \quad \beta_{2K}(\alpha) \geq \frac{25^{-2K}}{4}.$$

The profile bound gives $N \leq 7200K^3 25^{2K}$ at zero error. Since $K^3 \leq 8^K$ for integer $K \geq 1$, it also gives $N \leq 7200 5000^K$. This proves (6). $\square$

The occupation conclusion contains more information than a peak bound. For example, at the golden angle and zero error there are at most $16200k^5$ command epochs with at most k available labels, irrespective of the widths at the other epochs. The analogous rational-angle count is at most $7200k^3 25^{2k}$. These are proof bounds, not exact finite complexity tables.

4.1. An exact upper construction retained from the preceding argument. For completeness we give the construction needed for the upper side of (5); its role here is inherited [17]. A convergent p/q of α satisfies $|\alpha - p/q| < q^{-2}$. Choose one with $q \geq \max\{5, (40N)^{1/3}\}$ and put

$$a = \pi/q, \quad \delta = 2\pi(\alpha - p/q), \quad \lambda = \frac{\cos(a - |\delta|)}{\cos a}, \quad \omega = \frac{\sin |\delta|}{\lambda \sin(2a)}.$$

Here $|\delta| < 2\pi/q^2 < a$, so $\lambda \geq 1$ and $0 \leq \omega \leq 1$. Write $u_s = (\cos(2\pi s/q), \sin(2\pi s/q))$ for the regular polygon vertices, with subscripts modulo q .

Proposition 4.2 (Resonant stochastic polygon). *There is a q -state exact realization of (1), with command matrices independent of the epoch and a horizon-dependent final decoder. For bounded partial quotients it gives $W_N = O_\alpha(N^{1/3})$. For the golden angle the upper bound in (5) holds.*

Proof. At cut t use the vector $r_t u_s$ for label s , where $r_t = \lambda^t/4$. The idle row is

$$T_0(s, \cdot) = \lambda^{-1} \delta_s + (1 - \lambda^{-1}) \text{Unif}[q].$$

The active row is

$$T_1(s, \cdot) = (1 - \omega) \delta_{s+p} + \omega \delta_{s+p+\text{sgn } \delta}.$$

If $\delta = 0$ use $\omega = 0$ and omit the second term. The line joining u_{s+p} to its indicated neighbor intersects the ray at angle $2\pi s/q + 2\pi\alpha$ at radius $1/\lambda$. Resolving its two coordinates, or using the sine rule, gives exactly the displayed ω . Consequently the expected next vector is $r_t u_s$ after an idle command and $R_\alpha r_t u_s$ after an active command.

Since $\tan(\pi/q) \leq 2\pi/q$ for $q \geq 5$,

$$\log \lambda = \int_{a-|\delta|}^a \tan v \, dv \leq |\delta| \tan a < 4\pi^2/q^3 < 40/q^3.$$

Thus $r_N \leq e/4 < 1$. The polygon at $t = 0$ has inradius $\cos(\pi/q)/4 > \sqrt{2}/10$ and therefore contains all four seed vectors $x/10$. Initialize by a convex representation of the seed in that polygon. The final decoder uses mean $r_N e_j^T u_s$, which lies in $[-1, 1]$. Induction on the commands proves exactness for every word.

Consecutive convergent denominators have bounded ratios at bounded-type angles, yielding the claimed order. At the golden angle the denominators are Fibonacci numbers, and the first one not smaller than a threshold $L \geq 5$ is strictly less than $2L$. This gives the stated numerical bound. $\square$

For the rational rotation an elementary exact rational alternative retains a triangle label and the number of active commands. Initialize in the triangle from Proposition 2.2, increment the count only on the active command, and decode the rotated triangle vertex. It uses at most $3(N+1)$ labels and rational rows. The preceding resonant construction uses exact real rows; a finite fair-bit implementation of those rows is not being asserted.

5. DEFICIENCY OF A STOCHASTIC LIFT DIAGRAM

We next turn the quantitative converse into an obstruction to approximate stochastic lifts of the main experiment. The local language is classical comparison of experiments [3, 12]; the new use is a uniform lower bound on a full diagram, derived from the Fourier budget.

5.1. Finite garbling and its normalization. Let $E \in \mathbb{R}_+^{H \times K}$ and $F \in \mathbb{R}_+^{H \times J}$ have stochastic rows, indexed by a common finite parameter set. With the convention $\text{TV}(p, q) = \frac{1}{2} \sum_j |p_j - q_j|$, define

$$(19) \quad \delta(E, F) = \min_{T \mathbf{1} = \mathbf{1}, T \geq 0} \max_{h \in H} \text{TV}((ET)_h, F_h).$$

The direction is fixed: the signal of E is to simulate the signal of F . This is a one-sided finite deficiency. It is not a distance unless both directions and appropriate equivalence classes are considered.

Proposition 5.1 (Finite comparison formula). *The minimum in (19) is attained, and*

$$(20) \quad \delta(E, F) = \max_{\sum_h \text{osc}(Z_h) \leq 1} \left\{ \langle Z, F \rangle - \sum_{i=1}^K \max_{1 \leq j \leq J} (E^T Z)_{ij} \right\}.$$

Here $\text{osc}(Z_h) = \max_j Z_{hj} - \min_j Z_{hj}$; adding a constant to a row of Z changes neither objective nor constraint. In particular, $\delta(E, F) = 0$ if and only if $F = ET$ for a stochastic T , equivalently if and only if the associated finite decision inequalities hold. For three finite experiments,

$$(21) \quad \delta(E, G) \leq \delta(E, F) + \delta(F, G).$$

Proof. The stochastic matrices form a compact polytope. If each row of A sums to zero, then

$$\max_h \frac{1}{2} \|A_h\|_1 = \max_{\sum_h \text{osc}(Z_h) \leq 1} \langle Z, A \rangle.$$

Indeed each row contributes at most $\text{osc}(Z_h) \|A_h\|_1 / 2$; equality follows by using an indicator of the positive coordinates on a maximizing row. Subtract the minimum of each row of Z . The permitted Z then form the compact convex set specified by auxiliary variables $u_h \geq 0$, $0 \leq Z_{hj} \leq u_h$, $\sum_h u_h \leq 1$. Apply finite convex minimax to $\langle Z, F - ET \rangle$. Each row of T maximizes a linear form independently, giving $\max_T \langle Z, ET \rangle = \sum_i \max_j (E^T Z)_{ij}$. This proves (20), including the normalization of the dual budget. At zero error it is exactly the finite Blackwell garbling criterion, not a new comparison theorem.

For the triangle bound choose minimizing post-processings T and U . For every row, contraction of TV under U and the triangle inequality give

$$\text{TV}((ETU)_h, G_h) \leq \text{TV}((ET)_h, F_h) + \text{TV}((FU)_h, G_h).$$

Taking maxima proves (21). □

5.2. One common transition for every prefix. Let H_t be all seed/command prefixes of length t in (1). A finite proposed diagram $\mathcal{E}$ assigns a row-stochastic matrix $E_t \in \mathbb{R}_+^{H_t \times K_t}$ at every cut. These are proposed ideal hidden-state distributions. For $c \in \{0, 1\}$ let $E_{t+1}^{[c]}$ have rows $E_{t+1}(hc, \cdot)$ indexed by H_t . Let Q_j have rows equal to the prescribed binary answer laws for query j and complete prefixes in H_N . Define

$$(22) \quad \begin{aligned} d_t(\mathcal{E}) &= \max_{c \in \{0,1\}} \delta(E_t, E_{t+1}^{[c]}), \\ d_{\text{out}}(\mathcal{E}) &= \max_{j \in \{1,2\}} \delta(E_N, Q_j), \\ \mathcal{D}(\mathcal{E}) &= \sum_{t < N} d_t(\mathcal{E}) + d_{\text{out}}(\mathcal{E}). \end{aligned}$$

A stochastic post-processing in each term is common to all prefixes; it cannot depend on an unrecorded history. Each fixed-diagram term is a finite linear program. The definition does not optimize over the unknown rows E_t and does not assert that local errors cannot cancel at the output.

Theorem 5.2 (Executable composition and main-family defect). *For every diagram $\mathcal{E}$ there is a clocked stochastic machine with its same profile and uniform output error at most $\min\{1, \mathcal{D}(\mathcal{E})\}$. If every $K_t \leq K$, then*

$$(23) \quad \mathcal{D}(\mathcal{E}) \geq \frac{1}{2} \left[\frac{1}{10} - \sqrt{\frac{18K^3}{N\beta_{2K}(\alpha)}} \right]_+.$$

More generally, for $J_k = \#\{t < N : K_t \leq k\} > 0$, the same inequality holds with K^3/N replaced by k^3/J_k .

Proof. Initialize using E_0 and use minimizing stochastic matrices in each term of (22). Denote the actual prefix-state law by $\hat{E}_t(h)$. Induction and TV contraction give

$$\max_{h \in H_t} \text{TV}(\hat{E}_t(h), E_t(h)) \leq \sum_{u < t} d_u(\mathcal{E}).$$

The selected terminal decoder adds at most d_{out} . This proves the constructive statement, including a common executable row at every step.

Write $D = \mathcal{D}(\mathcal{E})$. If $D \geq 1/20$, the lower bound is immediate. Otherwise the constructed machine has error at most D , so Theorem 1.1 applies with $\kappa = 1/10 - 2D$. It implies $N\beta_{2K}(\alpha) \leq 18K^3/(1/10 - 2D)^2$. Rearranging gives (23). The occupation form is identical, using (3) rather than its peak consequence. $\square$

For badly approximable α , the right side is at least

$$(24) \quad \frac{1}{2} \left[\frac{1}{10} - \sqrt{\frac{18K^5}{b_\alpha^2 N}} \right]_+.$$

Thus no family of diagrams with $K = o(N^{1/5})$ has total certified simulation defect tending to zero. In particular a static polygon lift using $O(\log N)$ coordinates cannot be turned into a vanishing-defect stochastic diagram for this task at the same width. This conclusion requires a specified initialization and bounded terminal decoder as in (22); it is not a lower bound on static extension complexity. It permits every hidden choice of section and projection, not merely a selected regular-polygon extended formulation.

The additive quantity $\mathcal{D}$ is a useful operational upper bound on terminal error, not a claimed exact formula for minimum terminal error. Approximation of intermediate lifts, terminal decoding and the complete wordwise task are thereby kept distinct. Theorem 5.2 provides the requested

connection between finite deficiency and the actual rotation family without relying on a separate toy example.

6. EQUIVARIANCE AND TWO AUTONOMOUS SPECIFICATIONS

An equivariant lift carries a group action on one extension, intertwining with the action on the base. In the orthant case this is represented by coordinate permutations after the usual normalization. Definitions 2.7–2.8 and Theorem 2.9 of the preprint of Gouveia–Parrilo–Thomas [7] relate symmetric cone lifts to symmetric slack factorizations. Symmetry can change extension size substantially [9]. In the regular-polygon case, equivariant LP lifts of prime-power order require linear size, whereas nonsymmetric LP lifts can have logarithmic size; the equivariant PSD theory has different bounds [4], including logarithmic-size lifts of regular 2^n -gons.

Our stochastic diagrams do not impose a group representation, inverses, one common section, or a permutation action on hidden coordinates. Their maps may be noninvertible and epoch dependent. Hence the cited equivariant lower bounds do not apply directly, and static small lifts do not provide the stochastic maps needed here. The following elementary comparison makes one exact point of contact. It is a consequence of classical stochastic matrix spectral theory, not a new symmetric-lift bound.

Proposition 6.1 (An exact cyclic stochastic lift). *Let $\xi = e^{2\pi i/m}$, $m \geq 2$, and suppose stochastic rows $E_0, \dots, E_{m-1} \in \Delta_K$, a stochastic matrix T , and a complex column d satisfy*

$$E_j T = E_{j+1 \bmod m}, \quad E_j d = r \xi^j \quad (r \neq 0).$$

Then $K \geq m$. The bound is attained by the m -cycle permutation matrix.

Proof. The complex row $w = m^{-1} \sum_{j=0}^{m-1} \xi^{-j} E_j$ has $wd = r$, so it is nonzero, and direct reindexing gives $wT = \xi w$. Thus ξ is an eigenvalue of T . Every unit-modulus eigenvalue of a finite stochastic matrix belongs to a recurrent irreducible class and is a root of unity of order dividing the period of that class. That period is at most the number of its states. For completeness, transient blocks have spectral radius less than one; in an irreducible recurrent block, equality in the triangle inequality for an eigenvector at its maximal modulus forces its phases to advance by the eigenvalue along every positive edge. Along every directed cycle the accumulated phase is one, so its order divides the greatest common divisor of the cycle lengths. The primitive m th root therefore requires at least m states. The permutation construction realizes equality. $\square$

Proposition 6.1 assumes exact cyclic closure, including its last-to-first transition. The finite irrational experiment has no such closure. Its nonstationary lower bound comes instead from the finite Fourier occupation theorem, which needs no eigenvalue shared by successive matrices. This is why replacing the latter proof by a peripheral-spectrum argument would be invalid.

6.1. Fixed length is not an anytime query. Define A_N^{fix} by requiring one common state set, two common command matrices T_0, T_1 , and one final decoder, but correctness only after exactly N commands. The entire machine may depend on N . Define A_N^{any} with the same data required to answer after every command length $0, 1, \dots, N$. A query is an atomic external input in both models. Counts refer to internal labels before emission; charging two additional absorbing answer labels increases the given upper bounds by at most two. No internal test of input length is required by either interface.

Corollary 6.2 (Bounds with the query-time quantifier fixed). *For bounded-type α ,*

$$\Omega_\alpha(N^{1/5}) \leq W_N(\alpha) \leq A_N^{\text{fix}}(\alpha) \leq O_\alpha(N^{1/3}).$$

For irrational α , the stronger anytime specification satisfies

$$N + 1 \leq A_N^{\text{any}}(\alpha) \leq 3(N + 1).$$

The anytime assertion is the inherited spectral comparison [17], not a linear lower bound for A_N^{fix} .

Proof. The unrestricted lower bound applies to stationary matrices in particular. Proposition 4.2 uses the same command matrices at all epochs, so it gives the fixed-length autonomous upper bound without phase tags. Its final radius and decoder depend on N , as permitted.

For the anytime lower bound fix seed $(1, 1)$, use only active commands, and combine the two decoders into d . If a machine with $K \leq N$ states were correct, then $ET_1^j d = c\zeta^j$ for $j = 0, \dots, K$, where $c = (1 + i)/10 \neq 0$. Cayley–Hamilton, applied to these $K + 1$ identities, forces the characteristic polynomial of T_1 to vanish at ζ . The unit-modulus eigenvalue property proved above contradicts irrationality. For the upper bound keep the triangle label of the seed and the number of active commands. The latter never exceeds N on a permitted word. Idle commands leave it unchanged; active commands increment it; the decoder rotates the stored triangle vertex by that count. Arbitrary normalized rows at count N complete the matrix on unused transitions. This machine answers correctly at every allowed length. $\square$

7. RELATION TO PRIOR MECHANISMS AND REMAINING QUESTIONS

Positive realization has long distinguished observable linear dimension from positive-state dimension. Benvenuti–Farina [1, 2] give fixed-degree examples requiring large positive order and explain invariant-cone constructions. Heller [8] and the realization survey of Vidyasagar [14] provide the stochastic-process background. Accessible spaces, observable quotients and positive intertwining are also explicit in Monràs–Winter [11]. None of these mechanisms is renamed here as a new normal form. The specific assertion proved in Theorem 3.5 is a finite-horizon lower bound for unrelated stochastic updates, using a final correlation rather than one repeated transition or a statewise predictive coordinate.

For comparison, the recent work of Lumbreras–Ma–Thompson–Gu [10] studies irreducible finite-classical prediction and decision errors in a stationary world-model setting. Its Section IV.B uses repeated application of the same update along a Wait–Tick trajectory and the peripheral behavior of that update. Here the matrices may change at every epoch, and even the complete machine may change with the horizon. The phase/Jensen/Fejér proof gives a quantitative finite budget under those quantifiers. This distinction does not claim that all neighboring approximate-realization results have been exhausted.

Static polyhedral lift theory identifies extension size through slack factorizations [7]. Regular polygons admit small nonsymmetric extensions [5, 13]; respecting a group action can change this substantially [9, 4]. Section 6 states the exact model distinction. Our converse does not assume equivariance or extract an exposed face for each state. A hidden register instead produces at most K atoms in a measure of conditional *past* phases, even when its reachable simplex section has exponentially many projected vertices. The common amplitude-loss estimate is what makes those atoms useful through time.

The one-step equation $ET = F$, its decision inequalities, and one-sided TV deficiency belong to classical comparison of experiments [3, 12]; see also the modern formulation of the Blackwell–Sherman–Stein theorem in [6]. Proposition 5.1 includes a proof to fix the direction and factor of two, not to claim a new one-step duality. The additive composition of stochastic errors is likewise standard. The main-family lower bound (23), which applies to every proposed small-width diagram, is a consequence of the new all-hidden-state budget.

Statement	Role in the present article
Conditional angular-loss and Fourier occupation theorem	New quantitative converse for arbitrary clocked hidden registers.
Bounded-type polynomial and rational-angle logarithmic lower bounds	Consequences for the original rotation family, including fixed row error.
Resonant cube-root exact realization	Inherited upper construction, reproduced with proof.
Finite Blackwell/deficiency dual and TV composition	Classical ingredients, with conventions and proof supplied.
Main-family diagram-defect inequality	New consequence combining the converse with executable composition.
Cyclic spectral and anytime comparisons	Classical spectral mechanism and inherited stronger-task comparison.

7.1. Conclusions. For the fixed-alphabet uniformly positive rotation experiment, a hidden register of bounded-type angle requires polynomial, not merely logarithmic, width. The proof permits arbitrary hidden states and arbitrary clocked updates. The same occupation theorem survives a fixed uniform row error below $1/20$ and obstructs every small-width stochastic lift diagram with vanishing total deficiency.

The exact hidden exponent is not determined: the present lower and upper orders are $N^{1/5}$ and $N^{1/3}$. The vector-state order remains $N^{1/3}$ under the inherited bounded-type hypotheses. Sharpening the homogeneous angular estimate, the finite Fourier witness, or the exact hidden construction are distinct possible routes to closing this gap. No optimal global lift-search algorithm is asserted by the finite comparison linear programs.

The model counts persistent classical labels, not construction time, read-only program size, or fair-bit sampling workspace. Fixed-length and anytime autonomous interfaces remain distinct. The historical geometric, analytic and operator-theoretic manuscripts in the broader program keep their independent proof obligations; no conclusion in this article asserts their closure. Earlier results are preserved separately without being used as evidence of priority or editorial significance.

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